

MODEL QUESTION PAPER

TED(10) 4107

(Revision 2010)

FIFTH SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY

YARN MANUFACTURE – III

Time : 3 hrs

(Maximum marks: 100)

PART - A

(Answer **all** questions in one or two sentences. Each questions carries 2 marks)

Marks

I.

- 1, State the function of eccentric load selector in SKF PK 225 top arm
- 2, Mention what are ought travelers
- 3, State the function of yarn break detector in Siro spinning
- 4, Mention the function of flyer in TFO
- 5, Express the count of yarn in English system if the length of 0.25 lbs of yarn is 8400 yards

(2x5 = 10)

PART – B

Answer any **five** questions. Each question carries 6 marks

II.

- 1 Draw the anti wedge ring , list its design features and explain the advantages.
- 2, Describe how variable spinning speed is achieved using variable speed motor
- 3, Mention the advantages and disadvantages of open end rotor yarn over ring spun Yarn
- 4, Describe the function and constructional details of opening roller in rotor spinning Machine
- 5, Compare between ring doubler and Two For One Twister
- 6, Express the need of stop motion in doubling and describe with sketch the working of thread stop motion
- 7, A ring frame employs 15 mm diameter empty bobbin. Minimum angle of yarn pull is 23 degree. Find out the maximum practical ring diameter

(6x5 = 30)

PART – C

(Answer **one full** question from each module. Each question carries 15 marks)

Module I

- III 1. Compare between S- Twist and Z - Twist (3).
2, Explain the need and functions of traversing thread guide , ABC ring and stepped nose bar in ring spinning frame (12)

OR

- IV 1 Express the advantages and limitations of elliptical traveler (3)
2. State the problems caused by spinning triangle and explain with sketch how it is eliminated using elite compact spinning (12)

Module II

- V. 1. State the function of take up roller in open end rotor spinning (3)
2. Draw sketches of S, V, T, & G groove rotors and explain its spinning characteristics (12)

OR

- VI. 1. Compare between True twist and False Twist (3)
2. Describe the working of DREF II friction spinner with a neat sketch and explain the yarn formation (12)

Module III

- VII. 1 List the objects of wet doubling (3)
2, Sketch and explain how twisting, winding , and building are carried out in a ring Doubler (12)

OR

- VIII. 1. State the need for Reeling of yarn (3)
2 Express how cross reeled yarn is produced showing the sketch of a reeling machine (12)

Module IV

IX.

1. Calculate the production per spindle per 8 hours in grams from the data given

Spindle speed - 17500rpm
Count spun - 100s
Twist Multiplier - 3
Efficiency - 90 %
(7)

- 2, Calculate the production per TF0 per 8 hours in Kgms from the details given

Spindle speed - 13250rpm
TM - 4.8
Counts doubled - 16s and 24s
No. of spindles - 240
Efficiency - 93 %
(8)

OR

- X. 1 Calculate the production per open end rotor spinner per 8 hours in kgs from the data given

Rotor speed - 115000 rpm
TM - 4.7
Count spun - 12s
Efficiency - 88 %
No. of rotors - 220
(7)

- 2 Find out the main draft wheel and break draft wheel required for the production of 40s on ring frame from a roving hank of 1.35 and a break draft of 1.2. Break draft constant 62, main draft constant 564
(8)

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